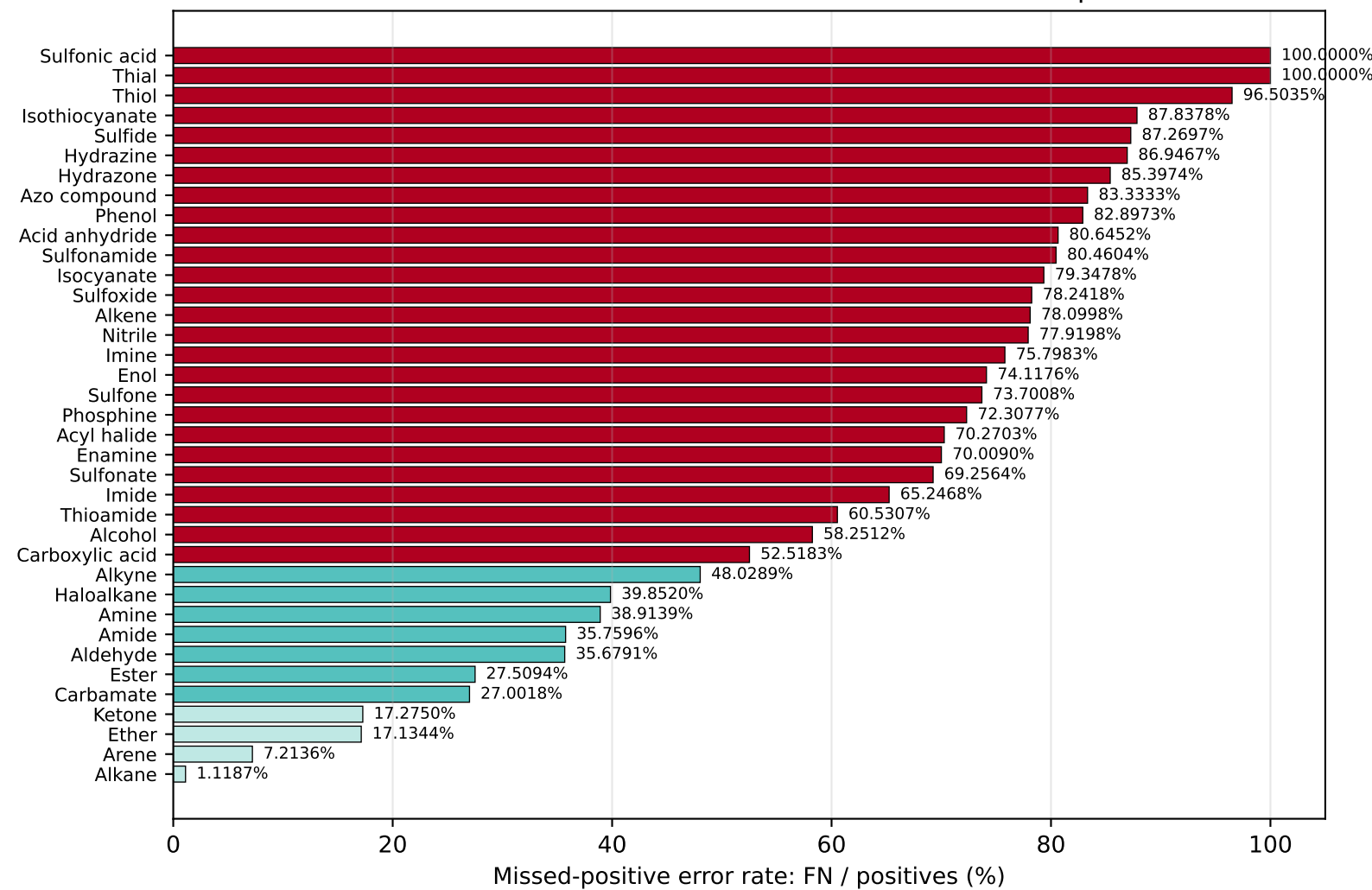
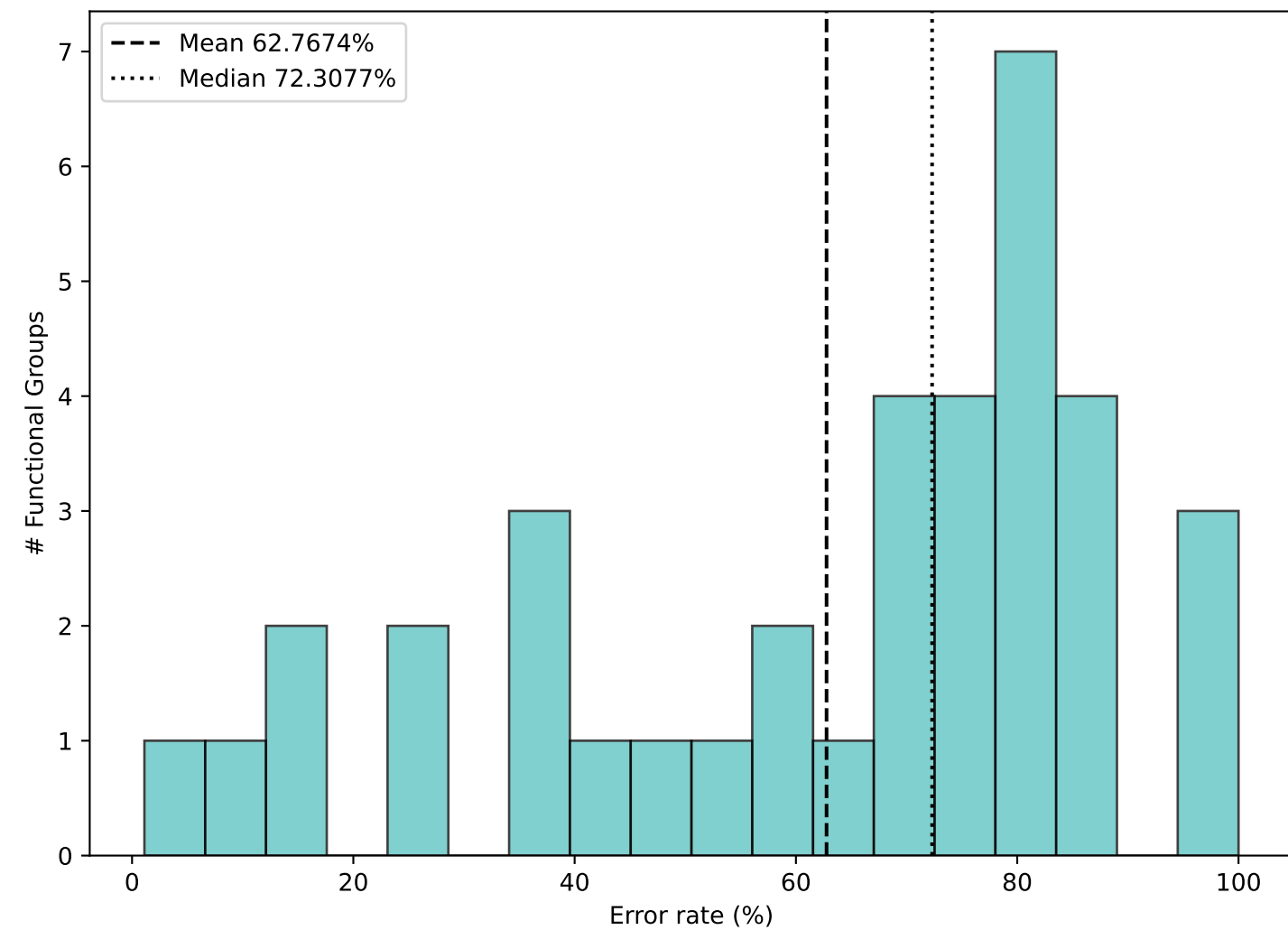


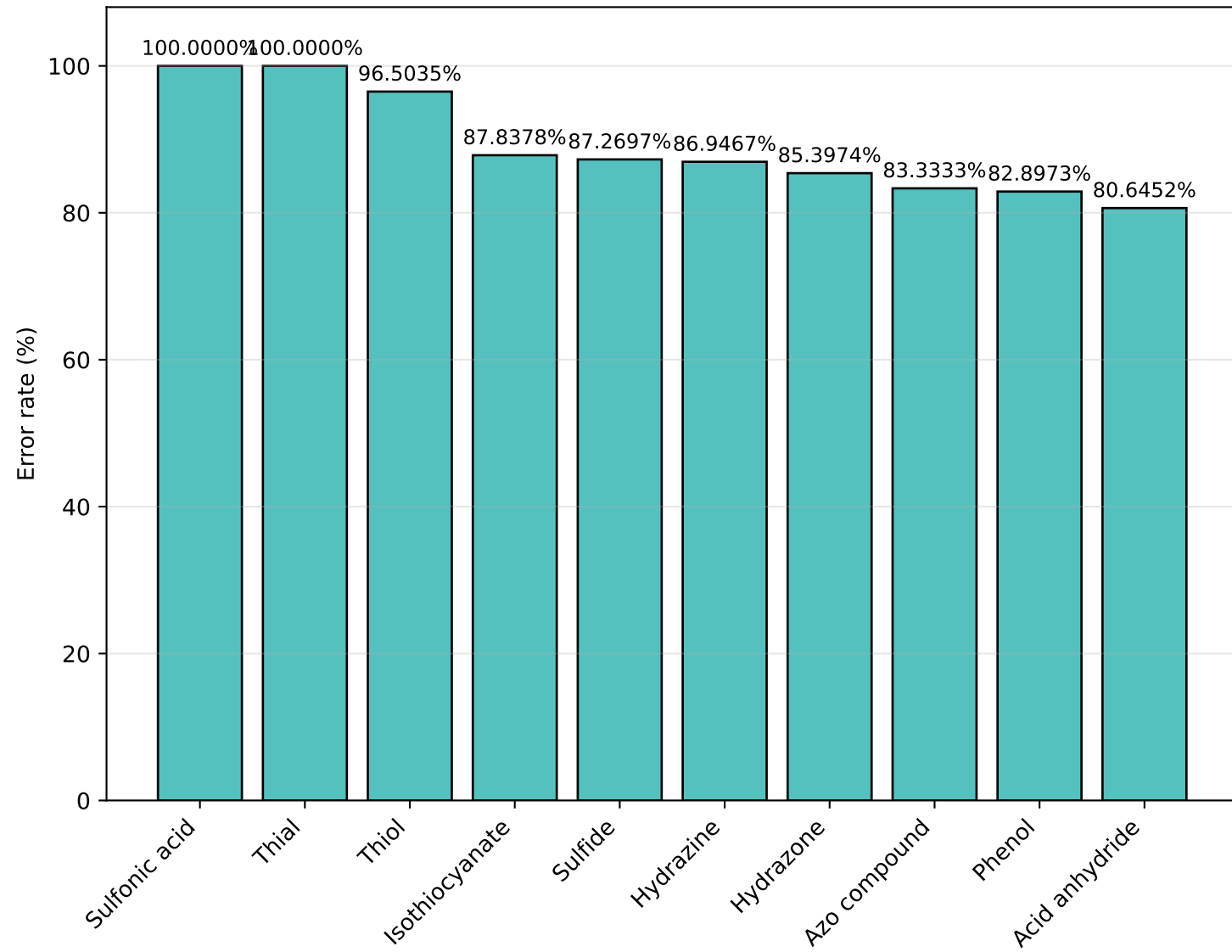
TTN C-NMR Validation Set - All 37 Functional Groups



Error-Rate Distribution



Top 10 Problem Groups



TTN C-NMR VALIDATION SET ERROR ANALYSIS

Global metrics:

F1 micro:	0.7150
F1 macro:	0.3290
Precision micro:	0.7149
Recall micro:	0.7150
Hamming accuracy:	0.9264

Error-rate statistics:

Mean:	62.7674%
Median:	72.3077%
Std:	26.7750%
Max:	100.0000% (Sulfonic acid)
Min:	1.1187% (Alkane)

Problem classes:

Severe (>50%):	26 groups
High (20-50%):	7 groups
Moderate (5-20%):	3 groups
Low (<=5%):	1 groups